

LUGOJ, Romania

WMO: 152700

Lat: 45.6867N

Lon: 21.9333E

Elev: 408

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.4	18.6	9.6	9.1	16.3	14.6	11.7	20.5	23.1	37.9	21.3	36.3	2.8	120	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.2	92.5	70.9	89.5	70.6	86.2	69.7	74.2	85.0	72.7	84.1	71.2	82.8	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.1	116.6	78.4	69.4	109.8	76.8	68.0	104.3	74.9	37.9	85.2	36.6	83.7	35.3	83.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	15.7	12.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	32.4	38.7	44.7	53.7	62.3	69.1	71.1	72.1	63.1	53.7	45.7	36.1	(6)
(7)		DBStd	15.37	8.72	7.95	7.38	8.49	5.98	5.82	5.43	5.77	7.66	8.03	8.49	7.71	(7)
(8)		HDD50	1800	548	325	194	61	2	0	0	0	1	57	180	432	(8)
(9)		HDD65	4847	1012	737	629	345	121	28	11	9	126	355	580	894	(9)
(10)		CDD50	3127	2	8	29	171	384	573	654	685	394	173	51	3	(10)
(11)		CDD65	701	0	0	0	6	39	151	201	229	69	6	0	0	(11)
(12)		CDH74	7341	0	0	3	121	450	1494	2086	2406	680	100	1	0	(12)
(13)		CDH80	2768	0	0	0	10	83	509	832	1097	226	11	0	0	(13)
(14)	Wind	WSAvg	4.4	4.3	5.3	4.9	5.0	4.4	4.1	4.1	3.8	3.9	3.8	5.3	4.3	(14)
(15)	Precipitation	PrecAvg	28.0	1.7	1.5	1.6	2.4	2.8	3.5	3.1	2.2	2.7	2.3	2.0	2.2	(15)
(16)		PrecMax	38.1	3.4	4.8	4.6	4.1	6.8	8.0	6.6	4.6	7.3	5.2	5.0	5.7	(16)
(17)		PrecMin	18.2	0.5	0.1	0.3	0.5	1.1	1.1	0.4	0.3	0.5	0.1	0.0	0.3	(17)
(18)		PrecStd	5.3	0.8	1.0	1.1	0.9	1.4	1.7	1.7	1.1	1.4	1.3	1.1	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	61.9	71.6	81.4	85.6	92.0	95.8	97.1	91.9	81.3	69.6	58.5	(19)
(20)			MCWB	49.3	51.9	55.7	62.7	66.2	73.3	70.7	70.7	68.9	63.3	56.8	50.4	(20)
(21)		2%	DB	50.0	57.7	66.2	77.7	82.7	88.9	92.1	93.1	85.5	76.4	64.8	53.1	(21)
(22)			MCWB	45.3	49.4	52.5	61.2	66.1	72.5	70.8	70.6	66.7	61.3	55.8	48.2	(22)
(23)		5%	DB	46.5	54.4	61.1	73.3	79.2	85.8	88.0	89.9	81.6	72.1	61.9	49.7	(23)
(24)			MCWB	43.5	47.0	50.3	58.1	64.3	71.0	70.4	70.2	65.5	60.2	54.3	46.0	(24)
(25)		10%	DB	43.6	50.6	56.8	69.2	75.9	82.7	84.3	86.1	77.2	67.2	58.1	45.7	(25)
(26)			MCWB	41.1	45.4	47.5	56.3	62.8	69.8	69.0	69.5	64.6	58.3	52.7	43.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	53.4	57.0	64.3	70.2	77.3	75.1	75.5	70.7	66.2	58.9	52.2	(27)
(28)			MCD B	55.7	61.2	70.8	77.7	81.9	87.8	84.5	88.5	87.5	76.7	64.9	55.6	(28)
(29)		2%	WB	46.4	50.4	53.6	62.2	67.9	74.6	73.5	73.5	68.6	63.1	56.9	48.9	(29)
(30)			MCD B	49.1	56.3	63.5	75.3	78.4	84.3	83.8	85.8	81.1	72.9	64.1	53.6	(30)
(31)		5%	WB	44.0	47.9	51.2	60.1	66.2	72.8	72.0	72.0	67.2	61.1	55.1	46.0	(31)
(32)			MCD B	46.2	52.8	59.5	72.1	76.0	83.2	83.8	84.2	77.7	69.7	61.3	49.1	(32)
(33)		10%	WB	40.7	45.8	49.0	57.4	64.2	70.8	70.6	70.6	65.7	59.2	53.3	43.8	(33)
(34)			MCD B	43.3	50.1	55.6	66.6	72.7	80.4	82.1	82.6	74.7	66.6	58.0	46.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	14.7	18.8	20.8	20.5	21.4	22.6	23.2	20.2	20.3	14.6	11.1	(35)
(36)			MCDBR	15.5	20.1	25.8	27.7	26.6	26.4	28.2	29.1	27.2	28.3	19.9	17.4	(36)
(37)			MCWBR	10.9	12.5	14.2	13.1	12.0	11.1	10.4	9.2	10.4	14.1	11.5	11.7	(37)
(38)		5% WB	MCD BR	14.4	17.1	22.2	24.6	23.2	23.8	23.7	25.2	23.1	25.1	17.9	16.6	(38)
(39)			MCWBR	10.6	11.4	12.9	12.0	11.7	11.6	10.1	9.7	9.7	13.1	10.5	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.362	0.402	0.459	0.445	0.437	0.451	0.461	0.429	0.388	0.356	0.334	(40)	
(41)		taud	2.332	2.253	2.199	2.059	2.160	2.225	2.191	2.160	2.214	2.324	2.364	2.358	(41)	
(42)		Ebn at Noon	239	253	259	254	262	265	259	251	247	242	229	225	(42)	
(43)		Edh at Noon	25	32	39	49	46	43	44	44	39	30	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	389	650	1031	1426	1717	1907	1892	1704	1199	827	489	314	(44)	
(45)		RadStd	45	87	137	145	157	168	151	163	129	90	49	44	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (65 neighbors)	+1.30	N/A	N/A	N/A	N/A	N/A	-236	-356	+230	+115

Nomenclature: See separate page